Group Project
DSE4212/QF4212, 2026-27, Semester 1

Instructions

- Please choose ONE of the projects: either Project 1 OR Project 2.
- At the end of the projects each group should hand in a zipped folder containing:
  1. A final report on the results (5–10 pages),
  2. Code that was used to obtain the results.
- The submission deadline for final report and code is 15 November 2026, 11.59 pm

Project 1

Data-Driven Credit Scoring Using Explainable Machine Learning

Objective

Develop and compare machine learning models to predict the default risk of borrowers, with a strong emphasis on model performance and interpretability using tools such as SHAP values.

Learning Goals

- Apply supervised learning models (e.g., Logistic Regression, Random Forests, Gradient Boosting) for the prediction task.
- Perform feature engineering and preprocessing for financial data.
- Use model explainability tools (e.g., SHAP, LIME) to interpret results.
- Evaluate model performance using classification metrics and assess fairness.

Dataset

- **German Credit Data (UCI Repository)**
  `https://archive.ics.uci.edu/ml/datasets/statlog+(german+credit+data)`

Project Components

(a) **Data Cleaning and EDA:** Handle missing values, outliers, and skewed variables.

(b) **Feature Engineering:** Construct features such as loan-to-income ratio, debt-to-income, employment categories.

(c) **Modeling:** Fit and tune logistic regression, random forests, and AadaBoost classifiers.

(d) **Model Evaluation:** Use AUC, precision, recall, confusion matrix.

(e) **Explainability:** Apply SHAP or LIME to explain the most influential features and predictions.

(f) **Fairness Analysis (Optional):** Analyze potential bias by demographic groups (e.g., gender, age).

(g) **Business Recommendations:** Provide actionable insights for loan approval policies.

Deliverables

- Jupyter Notebook with well-documented code.
- Final report (5– 10 pages) summarizing methods, results, and visualizations.

Project 2

Portfolio Optimization Beyond Markowitz Using Data Science Methods

Objective

Move beyond the classical mean–variance framework by applying machine learning and data-driven methods to construct and evaluate portfolios. Compare approaches such as risk-based allocations, regularized regressions with the traditional Markowitz efficient frontier.

Learning Goals

- Understand the limitations of mean–variance optimization (estimation error, instability of weights).
- Apply modern alternatives such as:
  - Risk-based methods (equal risk contribution, minimum-variance portfolios).
  - Machine learning for covariance/return forecasting (e.g., shrinkage estimators, Lasso/Ridge, random forests).
- Evaluate out-of-sample performance with realistic backtesting protocols.
- (Optional) Analyze robustness, transaction costs, and turnover.

Dataset

- Daily prices of a basket of liquid US equities from the Kaggle dataset *S&P 500 Stocks (daily updated)*: `https://www.kaggle.com/datasets/andrewmvd/sp-500-stocks`. Compute daily returns from the adjusted close prices.
- Alternatively: Fama–French factor datasets or ETF universes.

Project Components

(a) **Data Preparation and EDA:** Clean asset price data, compute returns, normalize frequencies (daily/monthly).

(b) **Benchmark (Classical):** Implement Markowitz mean–variance optimization (plug-in covariance, no shorting).

(c) **Alternative Approaches:**

- Regularized optimization (Lasso, Ridge, Elastic Net for weight stabilization).
- Risk-based allocations.
- ML-based forecasts (predict covariance/returns using shrinkage, tree-based models).

(d) **Backtesting Framework:**

- Rolling-window estimation of covariance/returns.
- Out-of-sample evaluation with realistic constraints (no future data leakage).

(e) **Performance Metrics:**

- Sharpe ratio, Sortino ratio, maximum drawdown.
- Stability of weights and robustness across regimes.

(f) **Comparison and Analysis:** Contrast ML-based portfolios with classical Markowitz; discuss robustness and interpretability.

Deliverables

- Jupyter Notebook with well-documented code.
- Final report (5–10 pages) summarizing methods, results, and visualizations of efficient frontiers, backtest equity curves, and risk-return tradeoffs.

Final Report Guidelines & Assessment (Total: 30 points)

Purpose

Evaluate how clearly the project is framed, how appropriately methods are chosen and executed, the quality of results and discussion, and how effectively the written report communicates findings.

Suggested Structure

1. Overview of the problem
2. Dataset description
3. Initial data analysis / visualisation (IDA)
4. Machine learning methods used
5. Model performance evaluation & application results
6. Conclusion (incl. limitations & future work)
7. Appendix (as needed for extra figures, tables, details)

Format

The report should be formatted using **11 pt** font size, **1.5** line spacing, and standard **A4** page size.

Note. Projects need not "succeed" to receive high marks: the quality of methods, reasoning, and communication is central.